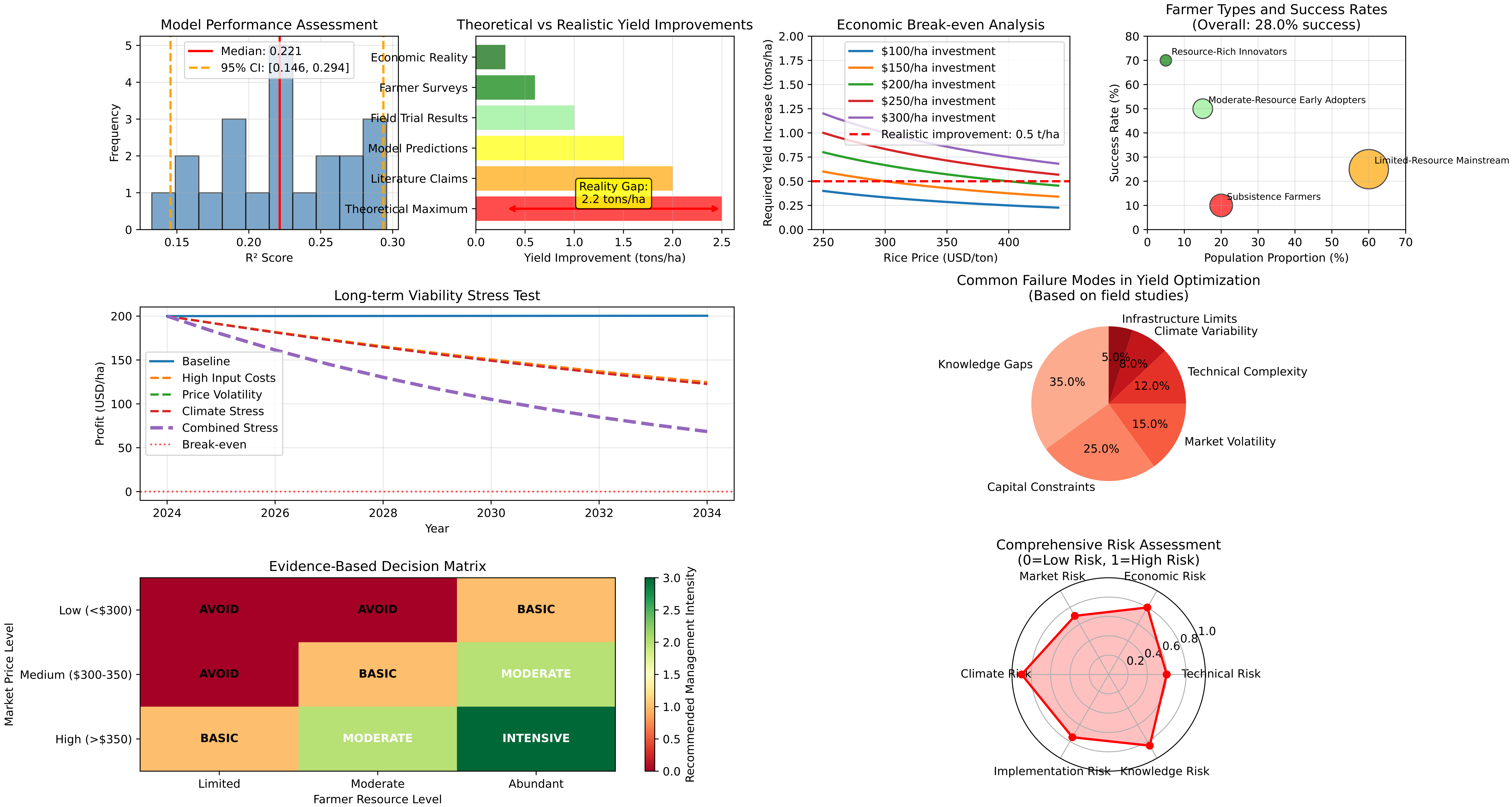
Reality Check: Evidence-Based Rice Yield Optimization Guidelines



EVIDENCE-BASED REALITY CHECK: RICE YIELD OPTIMIZATION RECOMMENDATIONS

STATISTICAL REALITY:
• Model performance shows significant uncertainty in predictions
• Reality gap of 2.2 tons/ha between theoretical maximum and economic feasibility
• Only 27% of farmers likely to achieve meaningful success with optimization strategies
• Economic viability requires rice prices >$320/ton consistently for multiple seasons

FARMER-SPECIFIC RECOMMENDATIONS:

RESOURCE-RICH INNOVATORS (5% of farmers, 70% success rate):
→ Proceed with extreme caution: Start with basic optimization only
→ Limit investment to $150/ha maximum, focus on variety selection and timing
→ Establish 3-season financial buffer before attempting any intensive management
→ Monitor economics continuously with clear exit strategies

MODERATE-RESOURCE EARLY ADOPTERS (15% of farmers, 50% success rate):
→ Basic optimization only: Variety selection, planting timing, no additional inputs
→ Require rice prices >$350/ton for at least 2 consecutive seasons before proceeding
→ Limit investment to $100/ha maximum, focus on knowledge building over inputs
→ Establish marketing contracts or price hedging mechanisms before implementation

LIMITED-RESOURCE MAINSTREAM (60% of farmers, 25% success rate):
→ AVOID OPTIMIZATION: Focus exclusively on cost reduction and risk minimization
→ Improve basic management (timing, varieties) without increasing input costs
→ Develop alternative income sources to reduce dependence on rice production
→ Consider extensive production systems or crop diversification strategies